

**ECE 341**  
**Homework 4**

The problem numbers listed below are from the Lathi textbook, 3<sup>rd</sup> edition.

| Prob. Number     | Course Outcome | Grade        |
|------------------|----------------|--------------|
| 4.1-1<br>(a) (d) | 1-c, 7-c       | / 20         |
| 4.1-4<br>(a)     | 1-c, 7-c       | / 10         |
| 4.2-2<br>(a) (b) | 1-c, 7-c       | / 20         |
| 4.3-1<br>(b)     | 1-c, 7-c       | / 15         |
| 4.4-7            | 1-c, 7-c       | / 20         |
| 4.8-2            | 1-c, 7-c       | / 15         |
| <b>TOTAL:</b>    |                | <b>/ 100</b> |

**1-c.** Use Laplace, Z, and Fourier transforms to obtain the frequency domain models to represent signals and systems.

**7-c.** Apply new knowledge to understand continuous and discrete systems.

- 4.1.1 a) First, we may split the given function into two integrals, on the interval where the given unit functions resolve to non-zero values, as

$$\int_0^{\infty} u(t)e^{-st} dt - \int_1^{\infty} u(t-1)e^{-st} dt = \int_0^{\infty} e^{-st} dt - \int_1^{\infty} e^{-st} dt.$$

Then, evaluating the integrals,

$$\begin{aligned} &= \left. \frac{-e^{-st}}{s} \right|_0^{\infty} + \left. \frac{e^{-st}}{s} \right|_1^{\infty} \\ &= \left[ \frac{-e^{-s\infty}}{s} + \frac{1}{s} \right] + \left[ \frac{e^{-s\infty}}{s} - \frac{e^{-s}}{s} \right] \\ &= \frac{1 - e^{-s}}{s}. \end{aligned}$$

The given function converges only when  $\sigma > 0$ , as the imaginary part of  $s$  will continue to oscillate indefinitely (an everlasting sinusoid). Therefore, we may assume that the ROC for this function is  $\Re(s) > 0$ .

- d) Again, splitting the given function into two integrals,

$$\int_0^{\infty} e^{-2t}e^{-st}u(t) dt - 2 \int_0^{\infty} e^{-t}e^{-st}u(t) dt = \int_0^{\infty} -e^{-t(s+2)} dt + 2 \int_0^{\infty} e^{-t(s+1)} dt.$$

Then, evaluating the integrals,

$$\begin{aligned} &= \left. -\frac{e^{-2t}e^{-st}}{s+2} \right|_0^{\infty} + \left. \frac{2e^{-t}e^{-st}}{s+1} \right|_0^{\infty} \\ &= \frac{1}{s+2} + \frac{2}{s+1}. \end{aligned}$$

The region of convergence for this function depends on both  $\sigma$  and  $j\omega$ , the real and imaginary parts of  $s$ . Under the condition where  $\sigma = -2$  and  $\omega = 0$ , the function will *not* converge, as well as under the condition where  $\sigma = -1$  and  $\omega = 0$ . Therefore, the function will converge as long as  $s \neq -2$  and  $s \neq -1$  (implying an imaginary part of 0 in both cases).

- 4.1.4 a) Using partial fraction decomposition,

$$\frac{2s+5}{s^2+5s+6} = \frac{2s+5}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

$$2s+5 = A(s+3) + B(s+2) = sA + 3A + sB + 2B$$

$$2s+5 = s(A+B) + 3A+2B.$$

Solving for  $A$  and  $B$ ,  $A+B=2$  and  $3A+2B=5$ , so

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

and so  $A=1$  and  $B=1$ .

Therefore,

$$\frac{2s+5}{s^2+5s+6} = \frac{1}{(s+2)} + \frac{1}{(s+3)},$$

and the inverse Laplace transform of the function may be found using Table 4.1, specifically the unilateral transform pair  $1/(s-\lambda) \Leftrightarrow e^{\lambda t}u(t)$ ,

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s+2} + \frac{2}{s+1}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\} \\ &= [e^{-2t} + e^{-3t}]u(t).\end{aligned}$$

- 4.2.2 a) Using the Laplace transform pair  $u(t) \Leftrightarrow 1/s$  and the time-shifting property of the Laplace transform,  $x(t-t_0) \Leftrightarrow X(s)e^{-st_0}$ ,

$$\begin{aligned}\mathcal{L}\{u(t) - u(t-1)\} &= \mathcal{L}\{u(t)\} - \mathcal{L}\{u(t-1)\} \\ &= \frac{1}{s} - \frac{e^{-s}}{s} = \frac{1-e^{-s}}{s}.\end{aligned}$$

- b) Likewise, using the time-shifting property of the Laplace transform along with the Laplace transform pair  $e^{\lambda t}u(t) \Leftrightarrow 1/(s-\lambda)$ ,

$$\mathcal{L}\left\{e^{-(t-\tau)}u(t-\tau)\right\} = \frac{e^{-s\tau}}{s+t}.$$

- 4.3.1 b) Converting to differential form,

$$\ddot{y}(t) + 4\dot{y}(t) + 4y(t) = \frac{d}{dt}e^{-t}u(t) + e^{-t}u(t)$$

with initial conditions

$$y(0^-) = \dot{y}(0^-) = 0,$$

the time differentiation property of the Laplace transform may be used along with the transform pair  $e^{\lambda t}u(t) \Leftrightarrow 1/(s-\lambda)$  to find

$$\begin{aligned}\mathcal{L}\{\ddot{y}(t) + 4\dot{y}(t) + 4y(t)\} &= \mathcal{L}\left\{\frac{d}{dt}e^{-t}u(t) + e^{-t}u(t)\right\} \\ &= [s^2Y(s) - sy(0^-) - \dot{y}(0^-)] + 4[sY(s) - y(0^-)] + 4Y(s) = sX(s) - x(0^-) + \frac{1}{s+1} \\ &= s^2Y(s) + 4sY(s) + 4Y(s) = \frac{s+1}{s+1} = 1 \\ &= Y(s)[s^2 + 4s + 4] = 1.\end{aligned}$$

Solving algebraically for  $Y(s)$ ,

$$Y(s) = \frac{1}{s^2 + 4s + 4} = \frac{1}{(s+2)^2},$$

which may be solved directly using the Laplace transform pair  $te^{\lambda t}u(t) \Leftrightarrow 1/(s-\lambda)^2$ , as

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\} = te^{-2t}u(t).$$

4.4.7 At  $t < 0$ , before the switch has opened, we may derive the initial conditions of the circuit as

$$\begin{aligned}v_s(0^-) &= 0^V \\ y(0^-) &= 10^A \\ y_2(0^-) &= 0^A.\end{aligned}$$

After the switch opens, at  $t > 0$ , we may derive the voltage across each component using the voltage equations for inductors, resistors, and capacitors,

$$\begin{aligned}v_R(t) &= Ri(t) \\ v_L(t) &= L \frac{di_L(t)}{dt} \\ v_C(t) &= \frac{1}{C} \int i(t) + v_0,\end{aligned}$$

along with the unilateral Laplace transform pairs

$$\begin{aligned}a &\Leftrightarrow \frac{a}{s} \\ x(t) &= X(s) \\ \frac{dx(t)}{dt} &\Leftrightarrow sX(s) + x(0^-) \\ \int_{0^-}^t x(\tau) d\tau &\Leftrightarrow \frac{1}{s} X(s)\end{aligned}$$

to derive the loop equation (KVL) for the circuit at  $t > 0$ ,

$$\begin{aligned}\frac{10}{s} &= 2sY(s) - 10 + sY(s) + \frac{1}{s}Y(s) + Y(s) \\ &= Y(s) \left[ 3s + \frac{1}{s} + 1 \right] - 10,\end{aligned}$$

so that

$$Y(s) = \frac{10s + 10}{3s^2 + s + 1}.$$

Transforming  $Y(s)$  back to the time domain *could* be done using partial fraction expansion, but a simpler method is to complete the square in the denominator and use the transform pairs

$$\begin{aligned}e^{-at} \cos btu(t) &\Leftrightarrow \frac{s + a}{(s + a)^2 + b^2} \\ e^{-at} \sin btu(t) &\Leftrightarrow \frac{b}{(s^2 + b^2)}\end{aligned}$$

. Mathematically,

$$\begin{aligned}Y(s) &= \frac{10}{3} \left[ \frac{(s + 1)}{(s^2 + s/3 + 1/3)} \right] \\ &= \frac{10}{3} \left[ \frac{s + 1/6 + 5/6}{(s + 1/6)^2 + 11/36} \right] \\ &= \frac{10}{3} \left[ \frac{s + 1/6}{(s + 1/6)^2} + \frac{5/6}{(s + 1/6)^2 + 11/36} \right].\end{aligned}$$

Then, using the tranform pairs above,

$$y(t) = \frac{10}{3}e^{\frac{-t}{6}} \left[ \cos\left(\frac{\sqrt{11}}{6}t\right) + \frac{5}{\sqrt{11}} \sin\left(\frac{\sqrt{11}}{6}t\right) \right] u(t) = y_1(t).$$

To find the voltage across the series inductor and capacitor, we may use their s-domain expressions, sum them, and convert back to the time domain. For their voltages,

$$V(s) = sY(s) + \frac{1}{s}Y(s).$$

Substituting  $Y(s)$ ,

$$V(s) = \frac{10s^2 + 10s}{3s^2 + s + 1} + \frac{10s + 10}{3s^3 + s + 1}.$$

Using similar methods as before to convert to the time domain,

$$v(t) = \frac{10}{3}\delta(t) + 10u(t) + e^{\frac{-t}{6}} \left[ \frac{-70}{9} \cos\left(\frac{\sqrt{11}}{6}t\right) + \frac{50}{9\sqrt{11}} \sin\left(\frac{\sqrt{11}}{6}t\right) \right] u(t).$$

4.8.2 Note that for the given transfer function  $H(s)$ , we may assume the system beign described is causal, since  $H(s)$  is *proper*, and so the system response  $Y(s)$  may also be assumed to be causal, applying only under the condition  $t > 0$ .

- a) Converting the input  $x(t) = 5 \cos(2t + 30^\circ)$  to the s-domain using a unilateral Laplace transform, based on the transform pair

$$re^{-at} \cos(bt + \theta)u(t) \Leftrightarrow \frac{(r \cos \theta)s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)},$$

we find

$$X(s) = \frac{5s \cos 30^\circ - 10 \sin 30^\circ}{s^2 + 2^2} = \frac{5s\sqrt{3}/2 - 5}{s^2 + 4}.$$

Multiplying the transfer function by the input in the s-domain (equivalent to convolution in the time domain), and expressing both  $H(s)$  and  $X(s)$  in root and pole form for simplicity,

$$Y(s) = \left[ \frac{(s+2)}{(s+1)(s+4)} \right] \left[ \frac{5(s\frac{\sqrt{3}}{2} - 1)}{(s+j2)(s-j2)} \right] = 5 \left[ \frac{0.866s^2 + 0.7321s - 2}{(s+1)(s+4)(s+j2)(s-j2)} \right].$$

- b) Once again, we will convert the input  $x(t) = 10 \sin(2t + 45^\circ)$  to the s-domain by first using the identity

$$\sin x = \cos x - 90^\circ$$

to avoid using a different transform pair than in part a), so that

$$x(t) = 10 \cos(2t - 45^\circ).$$

Converting to the s-domain,

$$X(s) = \frac{10s \cos -45^\circ - 20 \sin -45^\circ}{s^2 + 2^2} = \frac{10s/\sqrt{2} + 20/\sqrt{2}}{s^2 + 4}.$$

Multiplying by the transfer function to find  $Y(s)$ ,

$$Y(s) = \left[ \frac{(s+2)}{(s+1)(s+4)} \right] \left[ \frac{10(s+2)}{\sqrt{2}(s+j2)(s-j2)} \right] = 7.07 \left[ \frac{s^2 + 4s + 16}{(s+1)(s+4)(s+j2)(s-j2)} \right].$$

- c) Finally, we will convert the input  $x(t) = 10 \cos(3t + 40^\circ)$  to the s-domain using the same Laplace transform pair as before, as

$$X(s) = \frac{10s \cos 40^\circ - 30 \sin 40^\circ}{s^2 + 9} = \frac{7.66s - 19.28}{(s^2 + 9)}.$$

Then, multiplying by  $H(s)$  to find  $Y(s)$ ,

$$Y(s) = \left[ \frac{(s+2)}{(s+1)(s+4)} \right] \left[ \frac{7.66s - 19.28}{(s+j2)(s-j2)} \right] = 7.66 \left[ \frac{s^2 - 0.52s + 5.03}{(s+1)(s+4)(s+j2)(s-j2)} \right].$$